COMP47350 - Data Analytics (Conversion)
Data Quality Review: PPR Training Dataset (53,992 Transactions)

Executive Summary

The cleaned training dataset contains 53,992 unique residential property transactions (2016-2024) across 9 original raw transaction features, with 0% missingness in core fields (Price, County, Date). During processing, row_id and additional derived analytical features were created for validation and modelling. Routing-key missingness reduced from 68.70% to 32.60% (coverage increased from 31.30% to 67.40%; +36.10 percentage points) via address inference. Inference validation on known routing keys was based on 10,065 comparisons, with 349 mismatches, 3.47% error, and 96.53% accuracy. VAT Exclusive and IsSecondHand show a very strong practical association in cross-tab checks, with 0 observed rule violations. The dataset is suitable for modelling, with minor warnings on spatial granularity.

INTRODUCTION & SCOPE

Dataset: ppr-group-22312913-train.csv (raw: 54,000 → deduped: 53,992 rows)

Evidence pointer: main notebook, Step 1, Raw train shape before and after raw-signature dedup.

Period: 2016-01 to 2024-12 (2,479 unique dates)
Features: 9 original raw transaction features (Date, Address, County, Eircode, Price, Not Full Market Price, VAT Exclusive, Description of Property, Property Size Description).
Processing outputs also include row_id and multiple derived features used for DQR/DQP and modelling.
Target: Price (€) – 6,689 unique values, range €5,586.84–€111,280,172

Quality checks performed:

- Completeness: missingness by feature
- Accuracy: outliers, inference validation
- Consistency: VAT × property type crosstabs
- Validity: type conformity, constants
- Timeliness: temporal coverage

METHODOLOGY

Data Processing Pipeline (main.ipynb Steps 2-4):

1. Deduplication: 8 exact transaction duplicates removed
2. Parsing: Date (dd/mm/yyyy to datetime), Price (string to numeric euro value)
3. Categorical: County/flags → category dtype
4. Helpers: quality_overview(), numeric_summary()
5. Eircode: Routing key (first 3 chars) inference from 137-town reference
6. Validation: Inference accuracy on known routing keys

Key Decisions

- Address → cleaned (upper, alphanum only) for town matching
- Property desc → IsSecondHand boolean (TeachArasan→Second-Hand)
- No imputation beyond inference (preserve raw quality signals)
- Ambiguous routing keys (A82, A92) were excluded from county overwrite mapping to reduce erroneous county assignment.

Evidence pointer: main notebook, Step 5.1, Ambiguous routing keys excluded from Step 8 county overwrite output.

Reproducibility Note: Results are environment-dependent. In particular, pandas categorical handling can vary across machine setups, which may affect intermediate processing behaviour. To reproduce the reported values, run the notebook end-to-end in a single pinned environment and avoid mixing kernels or partial reruns.

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COMPLETENESS

Core Features: 100% Complete
- Price: 0 missing, 6,689 unique values ✓
- County: 0 missing, 26 Irish counties ✓
- Dates: 0 missing, 2,479 timestamps ✓
- Flags: 100% (Not Full Market Price, VAT Exclusive)

Evidence pointer: main notebook, Step 4 (Missing prices, Missing dates) and Step 2 quality overview (county completeness).

Mitigation: Routing-key missingness reduced from 68.70% to 32.60% (coverage 31.30% to 67.40%; +36.10 percentage points).
Evidence pointer: main notebook, Step 5.1, Missing routing key before inference and Routing-key missing after fill outputs.

ACCURACY

Price (€):
- Min: €5,586.84 | Max: €111,280,172 | Median: €250,000
- Distribution: Right-skewed (boxplot [Appendix Fig 1])
- Outliers: Valid (luxury/rural extremes)

Evidence pointer: main notebook, Step 4 (Min price, Max price) and Step 6 (target statistics and distribution plots).

Spatial Accuracy:
- County: Dublin 16,347 (30%), Cork 5,934

Evidence pointer: main notebook, Step 6, County frequency table.

- Eircode Routing: Inference validation on known routing keys used 10,065 comparisons, with 349 mismatches (3.47% error; 96.53% accuracy).

Evidence pointer: main notebook, Step 5.1, Compared rows, Mismatch rows, and Error rate outputs.

- Dublin mismatches dominant (142 cases, address parsing)

Evidence pointer: main notebook, Step 5.1, Top mismatch counties output.

Temporal Accuracy:
- Coverage: 2016-2024 complete ✓
- Eircode missingness by year: 99.7% (2016) → 24.5% (2024)

CONSISTENCY

VAT Exclusive × IsSecondHand:

Key Findings:
- New homes: 98.61% VAT=Yes ✓
- 2nd hand: 100% VAT=No ✓ (0 violations)
- Association strength: VAT Exclusive and IsSecondHand show a very strong practical association in cross-tab checks (0 violations).

Implication: Redundant features (drop VAT Exclusive).

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VALIDITY

Format Conformity:
- Price: All positive numeric ✓
- Dates: Valid datetime range ✓
- Flags: Binary (Yes/No) ✓
- No constants/duplicates post-cleaning ✓

Post-Conversion dtypes:
Date: datetime64 | Price: numeric dtype (float) | County: category ✓

TIMELINESS

Temporal Coverage: 2016-01 to 2024-12 (complete)
- 2,479 unique dates ✓
- Eircode adoption trend captured (99.7%→24.5% missing)

Current as of Dec 2024 ✓

Evidence pointer: main notebook, Step 5.1, Eircode missing by year output.

OVERALL ASSESSMENT

Strengths: Complete core features, perfect VAT consistency, full temporal span
Limitations: Eircode sparsity (mitigated), free-text sparsity

Data Quality Limitations: Address is free text and subject to spelling/format noise, which can affect routing-key inference and town matching. County-level geography remains coarse relative to local market variation, so some spatial heterogeneity is not captured. A residual share of routing keys remains unknown after inference, which limits granularity for some records.

Verdict: Suitable for ML – high quality target + geography ✓

RECOMMENDATIONS

Immediate (DQP):
1. Drop VAT Exclusive (redundant)
2. Use inferred routing keys (96.53% validation accuracy), with ambiguous A82 and A92 excluded from county overwrite.
3. Boolean IsSecondHand only

Future: External Eircode lookup for unmatched addresses

CONCLUSION

This Data Quality Review finds that the training dataset is fit for purpose for downstream property-price modelling. Core analytical fields are complete (Price, County, Date), duplicate transactions were removed transparently, and type conversions were applied consistently. Spatial quality was materially improved through routing-key inference, with missingness reduced from 68.70% to 32.60% and validation showing 3.47% error (96.53% accuracy) on known routing-key cases. Quality safeguards were explicitly applied, including exclusion of ambiguous routing keys (A82, A92) from county overwrite logic.

Overall, the dataset provides a reliable foundation for modelling, while the remaining limitations (free-text address noise, county-level granularity, and residual unknown routing keys) are documented and manageable within the Data Quality Plan.

Claim used in report	Exact value	Evidence pointer in notebook
Raw train rows before dedup	54,000	Step 10 run-all output, Cell 26: “Raw train shape (before raw-signature dedup)”
Exact duplicate rows removed	8	Step 10 run-all output, Cell 26: “Raw duplicate rows on full transaction signature”
Raw train rows after dedup	53,992	Step 10 run-all output, Cell 26: “Raw train shape (after raw-signature dedup)”
Price missingness	0	Step 4 quality checks output (also printed during Step 10 in Cell 26): “Missing prices”
Date missingness	0	Step 4 quality checks output (also printed during Step 10 in Cell 26): “Missing dates”
County missingness	0%	Step 2 quality overview table (printed during Step 10 in Cell 26)
Unique sale dates	2,479	Step 2 quality overview / date profile output (printed during Step 10 in Cell 26)
Unique price values	6,689	Step 2 quality overview table (printed during Step 10 in Cell 26)
Price minimum	€5,586.84	Step 4 output (Cell 26): “Min price”
Price maximum	€111,280,172.00	Step 4 output (Cell 26): “Max price”
Routing-key missing before inference	68.70%	Step 5.1 output (Cell 26): “Missing routing key before inference”
Routing-key missing after fill	32.60%	Step 5.1 output (Cell 26): “Routing-key missing after fill”
Routing coverage before inference	31.30%	Derived from Step 5.1 output: 100 − 68.70
Routing coverage after inference	67.40%	Derived from Step 5.1 output: 100 − 32.60
Routing coverage gain	+36.10 pp	Derived from Step 5.1 outputs: 67.40 − 31.30
Inference validation comparisons	10,065	Step 5.1 output (Cell 26): “Compared rows”
Inference mismatches	349	Step 5.1 output (Cell 26): “Mismatch rows”
Inference error rate	3.47%	Step 5.1 output (Cell 26): “Error rate”
Inference accuracy	96.53%	Derived from Step 5.1 output: 100 − 3.47
Dominant mismatch county	Dublin: 142	Step 5.1 output (Cell 26): “Top mismatch counties”
Largest county by count	Dublin: 16,347	Step 6 statistical report output (printed during Step 10, Cell 26): county frequency table
Second largest county by count	Cork: 5,934	Step 6 statistical report output (printed during Step 10, Cell 26): county frequency table
Earliest-year Eircode missingness	99.7% (2016)	Step 5.1 output (Cell 26): “Eircode missing % by year”
Latest-year Eircode missingness	24.5% (2024)	Step 5.1 output (Cell 26): “Eircode missing % by year”
Ambiguous routing keys excluded from county overwrite	A82, A92	Step 5.1 output (Cell 26): “Ambiguous routing keys excluded from Step 8 county overwrite”
Source of all DQR plots in appendix figures	One plotting block	Generated by Step 6 statistical_report, executed in Step 10 run-all (Cell 26)

Note: Unless otherwise stated, reported values are taken from the Step 10 full pipeline execution output (Cell 26), which executes Steps 1–9 in sequence and calls Step 6 for all DQR plots.